

Emergent Communication in Mixed-Motive Multi-Agent Reinforcement Learning: A Catan-Inspired Environment

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1 Introduction

Multi-Agent Reinforcement Learning (MARL) extends the traditional framework of Single-Agent Reinforcement Learning (SARL) to settings involving multiple interacting agents. While SARL has achieved significant success, supported by strong theoretical guarantees and practical algorithms such as Q-learning and SARSA, MARL introduces additional challenges.

A key assumption in SARL is the stationarity of the environment. In MARL, this assumption does not hold, as the learning process of each agent alters the environment from the perspective of others. This results in non-stationarity, often leading to instability and difficulty in convergence.

Another challenge arises from the exponential growth of the joint action space. For n agents each with a possible actions, the joint action space scales as a^n , making many classical approaches computationally infeasible.

An important and relatively underexplored aspect of MARL is the role of communication in environments that involve both cooperative and competitive interactions (mixed-motive settings). While prior work has examined communication in fully cooperative tasks, its impact in mixed settings remains less well understood.

2 Motivation

In many real-world multi-agent systems, agents must both compete and cooperate. These systems often require agents to:

- Compete for limited resources
- Cooperate temporarily for mutual benefit
- Exchange information or signals to coordinate actions

The board game *Settlers of Catan* provides a natural example of such a mixed-motive environment, where agents compete for victory while engaging in trade and negotiation.

This project proposes to use a simplified version of such an environment to study how communication influences coordination and performance in MARL.

3 Objectives

The primary objectives of this project are:

- To develop a simplified, reinforcement learning-friendly environment inspired by Settlers of Catan
- To implement a multi-agent reinforcement learning framework using self-play
- To investigate the role of communication in mixed cooperative-competitive settings

The central research question is:

How does communication influence coordination efficiency and agent performance in mixed-motive multi-agent environments?

4 Proposed Approach

4.1 Environment Design

A simplified “Catan-like” environment will be constructed with the following features:

- 2–3 agents interacting simultaneously
- A limited set of resource types
- Stochastic resource generation
- Simplified building and reward mechanisms
- An explicit trading mechanism between agents

The environment will be designed to be compatible with reinforcement learning frameworks and will focus on capturing the strategic importance of trading.

4.2 Learning Framework

A self-play multi-agent reinforcement learning setup will be used. Agents will share policy parameters while acting independently. A centralized critic with decentralized execution will be employed to improve training stability.

4.3 Communication Settings

Three communication settings will be studied:

1. **No Communication:** Agents act without any communication.
2. **Structured Communication:** Agents use a predefined set of messages corresponding to specific intents.
3. **Emergent Communication:** Agents learn to communicate using discrete message tokens without predefined semantics.

Communication will be constrained by a fixed bandwidth (limited number of tokens per timestep).

5 Evaluation Methodology

To assess the impact of communication, a structured evaluation protocol will be followed.

5.1 Within-Setting Evaluation

Agents trained under each communication setting will be evaluated in the same setting. Performance will be compared across:

- Win rate
- Average reward
- Trade success rate

This provides a direct measure of how communication affects performance.

5.2 Cross-Play Evaluation

Agents trained under different communication settings will be evaluated together in mixed populations. Examples include:

- Communication-enabled agents interacting with non-communicating agents
- Structured communication agents interacting with emergent communication agents

This evaluation helps assess:

- Robustness of learned strategies
- Dependence on shared communication protocols
- Generalization across agent populations

5.3 Communication Ablation

To verify whether communication is actively used by agents, ablation experiments will be conducted:

- Removing communication at evaluation time
- Replacing messages with random tokens

Performance degradation under these conditions indicates reliance on communication.

6 Analysis of Communication

Beyond performance, the structure and usefulness of communication will be analyzed using:

- **Message Entropy:** Measures diversity of communication
- **Consistency:** Whether messages correspond to consistent behaviors

This analysis helps determine whether communication is meaningful or merely incidental.

7 Ablation Studies

Additional ablations will be conducted to understand the role of communication design:

- Varying communication strength
- Introducing noise into communication channels
- Restricting or removing communication entirely

These studies help isolate the factors that influence communication effectiveness.

8 Expected Contributions

The project aims to contribute:

- A simplified and reproducible multi-agent environment for studying mixed-motive interactions
- A systematic empirical evaluation of communication in such environments
- Insights into the conditions under which communication improves coordination and performance

9 Conclusion

This work focuses on understanding the role of communication in multi-agent environments that combine competitive and cooperative elements. By studying communication under controlled settings and analyzing both performance and behavior, the project aims to provide insights into how communication emerges and influences learning in CATAN like simplified MARL systems.